GALOIS DESCENT FOR QUASI-COHERENT MODULES ON A FINITE ETALE SURJECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement opens with the author's own reference to the standing situation of the chapter ('as above'); that situation lies outside the counted span and enters only as context, while the statement and its proof are complete as written.

Editorial note. The counted spans carry 1 navigation stub; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- fpqc descent for quasi-coherent modules. Same chapter, Proposition proposition-fpqc-descent-quasi-coherent, invoked at proof lines 2092-2093. Supplies the equivalence of categories once the data are recognized as descent data.
- A surjective finite etale morphism is an fpqc covering. Standard, used at proof lines 2083-2084. Supplies that the descent theorem applies to the single morphism X to Y.
- The group action identifications with fibre products of X over Y. Standard, used at proof lines 2085-2089. Supplies the translation from group-indexed systems to descent data.

Source reference. Original proposition 023T: sources/stacks-descent/pinned-original-source.tex, source lines 1801–1801. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $X \to Y$, G, and $f_\sigma : X \to X$ be as above. The category of quasi-coherent $\mathcal{O}_Y$-modules is equivalent to the category of systems $(\mathcal{F}, (\varphi_\sigma)_{\sigma \in G})$ where*

(1) *$\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$-module,*
(2) *$\varphi_\sigma : \mathcal{F} \to f_\sigma^* \mathcal{F}$ is an isomorphism of modules,*
(3) *$\varphi_{\sigma\tau} = f_\sigma^* \varphi_\tau \circ \varphi_\sigma$ for all $\sigma, \tau \in G$.*

Proof. Since $X \to Y$ is surjective finite étale $\{X \to Y\}$ is an fpqc covering. Since $G \times X \to X \times_Y X$, $(\sigma, x) \mapsto (x, f_\sigma(x))$ is an isomorphism, we see that $G \times G \times X \to X \times_Y X \times_Y X$, $(\sigma, \tau, x) \mapsto (x, f_\sigma(x), f_{\sigma\tau}(x))$ is an isomorphism too. Using these identifications, the category of data as in the lemma is the same as the category of

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descent data for quasi-coherent sheaves for the covering $\{x \to Y\}$. Thus the lemma follows from Proposition 023T. $\square$